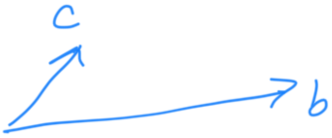
Cross product

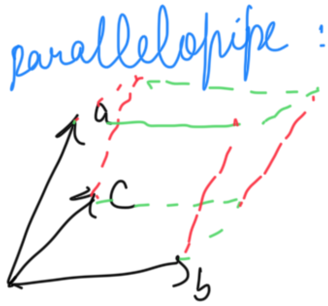
Area of ||gm



$= \vec{b} \times \vec{c}$

↳ vector orthogonal to $\vec{b}$ & $\vec{c}$

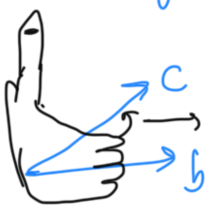
Volume of parallelopipe :



scalar triple product

is $|a \cdot (b \times c)|$

$b \times c$ → in dirn of b going to c counter clockwise



counter-clockwise

Properties

- $(a \times b) = -b \times a$
- $(\alpha \cdot a) \times b = \alpha \cdot (a \times b)$
- $\forall \vec{b}, \vec{c}$, there exists a unique vector r, s.t $a \cdot (b \times c) = a \cdot r$ $a \neq 0$
- $a \cdot (b \times c) = b \cdot (c \times a) = -a \cdot (c \times b)$
- $(a+b) \times c = (a \times c) + (b \times c)$
- $|a \times b| = |a||b| \sin\theta$
- $e_i \times e_i = 0 \;\; \forall i \in \{x, y, z\}$
- $e_x \times e_y = e_z, \; e_y \times e_z = e_x$

• $e_z \times e_x = e_y$

• $a \times b = \begin{vmatrix} e_x & e_y & e_z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$

• $a \cdot (b \times c) = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$

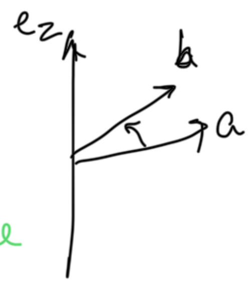
Note: $a = (x_1, y_1)$ $\hat{a} = (-y_1, x_1)$

For parallelogram with sides a & b,

area — +ve { rot^n from a to b from POV of e_z
— −ve is counter-Clockwise

$|\hat{a} \cdot b|$ → pseudo scalar product

or $|a \times b|$



diff. for this, take at facevalue rn.

• Cross Product = 0 → a & b are collinear

• Triple Product = 0 → a, b & c are coplanar

Diverging to true/pseudo vectors/scalars.

Parity Transformation

Inversion/reflection of spatial coordinates about origin.

In 3D space,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \xrightarrow{P.T} \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}$$

vectors
- true $\xrightarrow{P.T}$ sign change — $\vec{v} \xrightarrow{P.T} -\vec{v}$
- pseudo $\xrightarrow{P.T}$ invariant — $A \times B \xrightarrow{P.T} (-A \times -B)$

  $= A \times B$

Scalars
- true $\xrightarrow{P.T}$ invariant — $A \cdot B \xrightarrow{P.T} -A \cdot -B$

  $= A \cdot B$
- pseudo $\xrightarrow{P.T}$ sign change — $A \cdot (B \times C)$

  $\xrightarrow{P.T} (-A) \cdot (-B \times -C)$

  $= -A \cdot (B \times C)$

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line defined via $(r - r_0) \times d = 0$ { cross product property }

or

$(r - a) \times (b - a) = 0$

$d \rightarrow$ dir^n vector $\quad a, b \rightarrow$ 2 pts.

$r_0 \to$ init pt.

Similarly for planes: $(r-a)\cdot\left[(b-a)\times(c-a)\right] = 0$

or $(r-r_0)\cdot(d_1\times d_2) = 0$

$r_0 \to$ init pt.

$d_1, d_2 \to$ dir^n vectors

$a, b, c \to$ 3 pts.

In 2D, pseudo scalar product $\to$ +ve (if rot^n from $\vec{v_1}$ to $\vec{v_2}$ is clockwise)

$\to$ Can be used to calc area of polygons (disc. later)

Triple product is equivalent for 3D space.

Line Intersection

↳ Given two lines, To find: intersection points

let $L_1 \equiv r = a_1 + t d_1$

$d_1 \to$ dir^n

$a_1 \to$ init pt.

$t \to$ real parameter

$$L_2 \equiv (r - a_2) \times d_2 = 0$$

finding t:

$$(a_1 + t d_1 - a_2) \times d_2 = 0$$

$$\Rightarrow -(a_2 - a_1) \times d_2 + t(d_1 \times d_2) = 0$$

$$\Rightarrow \boxed{t = \frac{(a_2 - a_1) \times d_2}{d_1 \times d_2}}$$

Planes Intersection

$$\left.r \cdot n_i = a_i n_i \quad \forall \quad i \in \{1, 2, 3\}\right] \rightarrow \text{system of eqns.}$$

To Solve:

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (x_i\hat{i} + y_i\hat{j} + z_i\hat{k})$$
$$= (a_{ix}\hat{i} + a_{iy}\hat{j} + a_{iz}\hat{k}) \cdot (x_i\hat{i} + y_i\hat{j} + z_i\hat{k})$$

or

$$\underbrace{\begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}}_{} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\text{unknowns}} = \underbrace{\begin{bmatrix} a_{1x} x_1 + a_{1y} y_1 + a_{1z} z_1 \\ - - - - \\ - - - - \end{bmatrix}}_{B}$$

$A \rightarrow$ coefficient matrix $\quad (X) \quad$ vector/matrix of constants

By Cramer's Rule,

$AX = B$ where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

& $D = |A|$, we have

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}, \quad z = \frac{D_z}{D}$$

where D_x is det of matrix obtained by replacing 1st column of A with B

$D_y \rightarrow$ 2nd col by B & so on....

(Hence, we get a programmable soln.)